

Bio Informatics

Lecture 13

PAM Matrices

- Alignment scoring matrices are very useful in giving suitable scores to matches and mismatches.
- There are two types of scoring matrices in observed frequency measures. **PAMs** (**P**oint **A**ccepted **M**utations) and **BLOSUMs** (**B**LOCK **S**ubstitution **M**atrices).

PAM Matrices

Introduced by Margaret Dayhoff in 1978

Observed mutation probabilities between amino acids over 71 families of closely related proteins (85% sequence identity within a family)



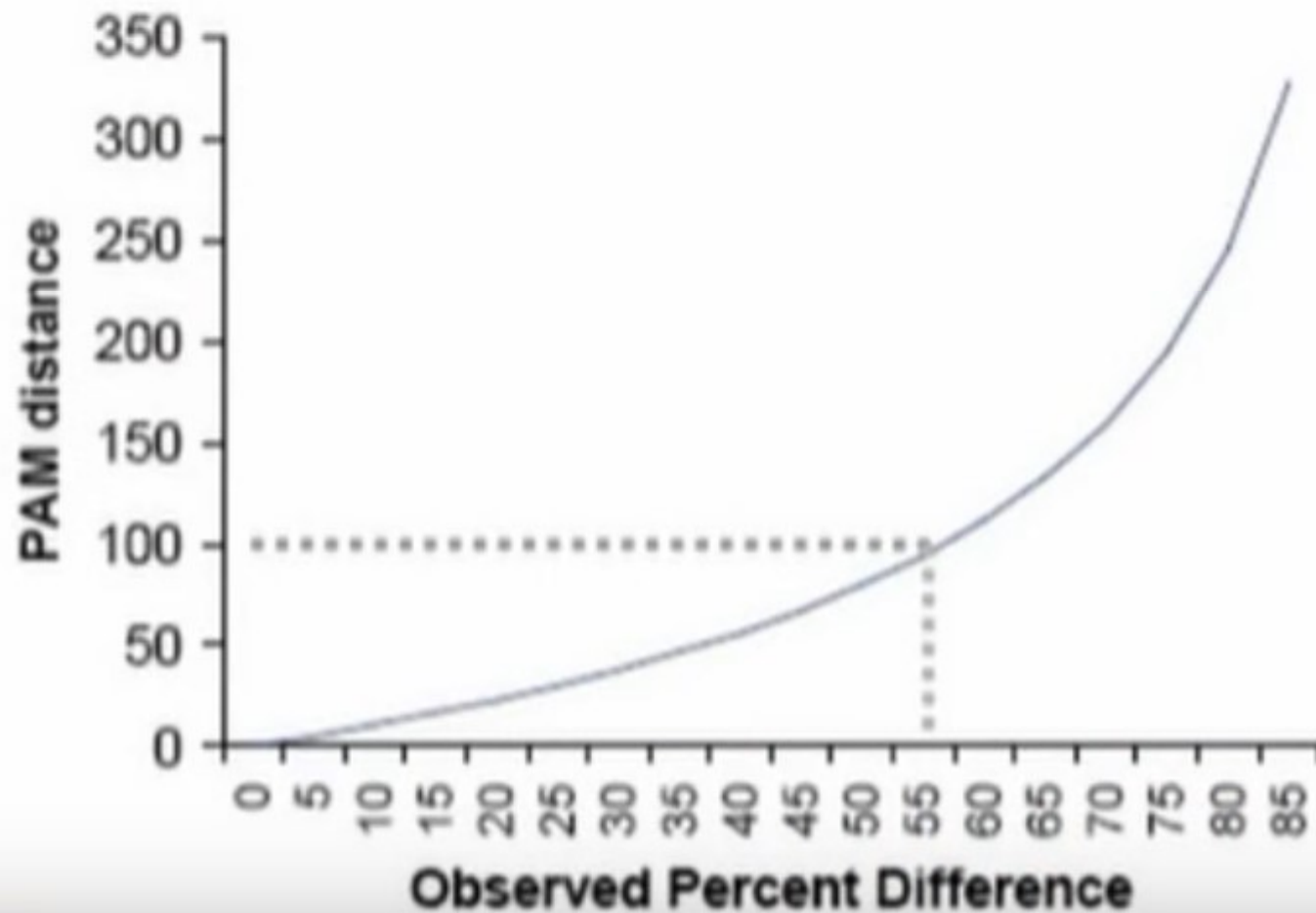
Based on a Markov mutation model; The PAM is a “unit of evolutionary mutation”. 1 PAM is the unit for which 1 mutation to occurs per 100 amino acids (this varies e.g. by species). The PAM₁ matrix express the log odds ratio of the likelihood of a point accepted mutation from one amino acid to another to the likelihood that these amino acids were aligned by chance.

PAM Matrices

- Point accepted mutation is a substitution/replacement of one amino acid by another such that the protein functions stay conserved.
- PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations.
- If two sequences diverge by 100 PAM units, it does not imply that they are totally different in each position.

PAM Matrices

Sequence Difference vs. PAM Distance



PAM Matrices

Steps to compute PAM matrices

Step 1: Align proteins sequences which are 1-PAM unit diverged

Step 2: Let $A_{i,j}$ be the number of times A_i is substituted by A_j

Step 3: Compute the frequency f_i of amino acid A_i

$$\text{Then, PAM1} = p_{ij} = \frac{A_{ij}}{\sum_k A_{ik}}$$

$$\text{PAM}'n' = (\text{PAM1})^n$$

PAM 250

Hydrophobic group: M, I, L, V
Aromatic group: F, Y, W.

PAM Matrices - Conclusion

- PAM matrices are scoring matrices that are useful in computing sequence alignment scores.
- PAM1, PAM2,...,PAM250 can be computed easily using aligned sequences of unknown related protein families.
- PAM120 is considered optimal scoring matrix among other PAMs.